

Waf: An Excellent Build Automation Tool

Build automation automates the process of a software build and removes the drudgery of having to do it all manually. Waf is a relatively new build automation tool that is open source and platform-independent. Written in Python, it is maintained by Thomas Nagy.



Build automation tools are used for automatic compilation and installation of computer software. Let us assume you are working on a large software project which is divided into a 100 program files. It is possible to compile all these files manually to generate an executable file. But what if you are frequently making changes to a particular file, and the functions used in it are also used by all the other program files? Then you'd have to recompile all the 100 files again and again. You'd be forced to execute hundreds of commands manually. Such a tedious process to compile the project might make you overlook some minor errors. In such a situation, a build automation tool is quite handy. It helps to build the executable file from hundreds of program files by automating the build process.

An executable file or a binary file contains instructions that cause a computer to perform certain tasks. We are familiar with the *.exe* executable files in Windows. The *a.out* file associated with C programs is actually an old executable

file format in Linux. And nowadays, the most popular executable file format in Linux is the Executable and Linkable Format (ELF).

Apache Ant, Bazel, BitBake, CMake, Make, Waf, etc, are some popular build automation tools. The most popular build automation tool is Make, which dates back to the 1970s. I used Make for a long time till I decided to explore Waf, a relatively new build automation tool, which I found to be excellent.

Installing Waf

Waf is developed using Python and maintained by Thomas Nagy. The latest version is *waf-1.9.6*. It is open source software and its source code is released under the new BSD licence. The associated documentation of Waf is under the Creative Commons Licence, and prohibits commercial reuse and modification. Hence, it is overlooked by many of the Linux distributions. For example, Debian Linux does not have Waf included in it by default. But having Waf in your